Leo Orshansky

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Pursuing a doctorate degree in theoretical computer science, specializing in quantum cryptography.

Education

2024 – Present **Ph.D. in Theoretical Computer Science, Columbia University**

2024 – 2026 **M.S. Computer Science, Columbia University**

2020 – 2024 **B.S. Computer Science (Turing Scholars Honors), University of Texas at Austin**
Honors thesis: *Computation in Isolated Thermodynamic Systems*.
B.S. Mathematics, University of Texas at Austin
Minor in Chinese Language, University of Texas at Austin

Research Experience

2024 – Present **PhD Research in Quantum Cryptography,** Columbia University

- *Doctoral Research, Advised by Profs. Tal Malkin and Henry Yuen:* Collaborating with researchers at Columbia, along with other institutions, to advance theoretical knowledge of quantum position verification, information-theoretic cryptography, secure multiparty computation, and other topics in the intersection of quantum computation and cryptography.

2022 – 2024 **Undergraduate Computer Science Research,** University of Texas at Austin

- *Undergraduate Honors Thesis, Supervised by Prof. David Soloveichik (Fall 2023):* Explored the connection between computational power of unconventional computing models (such as biochemical and reversible computing) and thermodynamic isolation. Co-wrote a research paper on a new framework for zero-energy computational sampling using the model of Brownian computation.

Research Publications

1. U. Girish, S. Goldwasser, G. Gluch, T. Malkin, **L. Orshansky**, and H. Yuen, "Private proofs of when and where," in *Advances in Cryptology – CRYPTO 2026*, To appear, 2026.
2. U. Girish, A. May, **L. Orshansky**, and C. Waddell, "Comparing classical and quantum conditional disclosure of secrets," *Quantum*, vol. 10, p. 2049, Apr. 2026, ISSN: 2521-327X. DOI: 10.22331/q-2026-04-01-2049.
3. Z. Deng, B. Ghaemmaghami, A. K. Singh, B. Cho, **L. Orshansky**, M. Erez, and M. Orshansky, "Enhancing cross-category learning in recommendation systems with multi-layer embedding training," in *Proceedings of the 15th Asian Conference on Machine Learning, PMLR 222*, PMLR, 2023, pp. 263–278.
4. D. Doty, N. Kornerup, A. Luchsinger, **L. Orshansky**, D. Soloveichik, and D. Woods, "Harvesting brownian motion: Zero energy computational sampling," Appeared as the Best Student Poster at DNA29 Conference, 2023. arXiv: 2309.06957 [cs.DS].

Employment Experience

2026 – 2027 — **Quantum Research Intern,** Hon Hai Research Institute (Foxconn). Remote
Researching various theoretical directions, primarily within the foundations of quantum cryptography (microcrypt). Supervised by Min-Hsiu Hsieh and collaborating with other members of the team, including Taiga Hiroka.

2024, Jun – Aug — **Software Engineering Intern,** Citadel Securities. New York, NY
Created a fully customizable, performant, and extensible alerting platform for options traders, built atop the highly performant KDB+ timeseries database and Q query language. Rolled out several high-priority alerts to traders, alongside a customization UI.

2023, Jun – Aug — **Software Engineering Intern,** Jane Street Capital. New York, NY
Added sidecar metrics such as per-symbol risk usage and tracking of inflight messages to the firm's primary order engine in OCaml and enabled publishing them to Prometheus + Grafana. Developed a new OCaml tool which enables root-privileged OCaml unit tests by transparently sandboxing them inside of QEMU virtual machines, and integrated this with the build system.

2022, Jun – Aug — **Software Engineering Intern,** Bloomberg LP. New York, NY
Rebuilt the "Mortgages Architecture Dispatcher" in Rust — a dynamically configurable reverse proxy providing transparent caching, UUID-based routing, and API versioning to the mortgages service mesh. Results: 99% reduction in dispatcher memory usage, and 50% speedup in cache hit latency.

2021, Jun – Aug — **Software Engineering Intern,** Cloudflare, Inc. Austin, TX
Designed and implemented an extension set for the Cloudflare Workers (serverless JS/WASM platform) Rust bindings. Deployed a Kubernetes application to host a new automated submission + grading system for the Cloudflare hiring assignment.

Awards and Achievements

2024 — **NSF Graduate Research Fellowship**, awarded by the National Science Foundation for the funding of research pursuits in theoretical computer science. Approx. $150,000 value.

Dean's Honored Graduate, College of Natural Sciences, UT Austin. This is the highest academic honor recognized by the college.

Honors Scholar with Research Distinction, College of Natural Sciences, UT Austin.

2023 — **Outstanding Chinese Language Student**, Department of Asian Studies, UT Austin.

2022 — **Distinguished Honors Scholar (Top 4%)**, College of Natural Sciences, UT Austin.

2020 — **Rocco C. Caffarelli Scholarship**, awarded on academic merit by Frost Bank. $20,000 value.

Salisbury Endowed Turing Scholarship, awarded on academic merit by University of Texas at Austin. $2,000 value.